

LaTeX 101

How to use LaTeX

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Contents

1	Section 1 - Basic Usage	2
1.1	Section 1.1 - Using Commands and Theorems	2
1.1.1	Propositions	2
1.1.2	Logical Operations (connectives)	3
1.3	Section 1.3 - Logical Equivalence	3
1.3.1	Testing for Equivalence using Truth Tables	4
1.4	Section 1.4 - Propositional Functions and Quantifiers	7
1.5	Section 1.5 - Nested Quantifiers	12
1.7	Section 1.7 Intro to Proofs	14
1.7.1	Set Notation	15
1.7.2	Methods of Proving $p \implies q$	16
2	Section 2 - Structures: Sets, Functions, Sequences, Sums, and Matrices	17
2.1	Section 2.1 - The Joy of Sets	17
2.2	Section 2.2 - Set Operations	21
2.3	Section 2.3 - Functions	25
2.4	Section 2.4 - Sequences and Series	33
2.6	Section 2.6 - Matrices	43
2.6.1	Properties of Matrices	47

Chapter 1

Section 1 - Basic Usage

1.1 Section 1.1 - Using Commands and Theorems

1.1.1 Propositions

Definition 1 (Proposition)

A **proposition** is a statement that is either

Example 1 (Examples with Propositions)

Determine if the following are propositions

1. It's all coming back to me now

Solution:

yes

2. Hello.

Solution:

No

3. $x + 3 = 10$

Solution:

No, since we don't know x

4. In a right triangle, $a^2 + b^2 = c^2$

Solution:

No, since a, b, c are not assigned anything

5. In a right triangle having legs of length a and b , and hypotenuse of c , then $a^2 + b^2 = c^2$

Solution:

Yes, since a, b, c are properly specified

6. There are 75 people in this room or the sky is purple

Solution:

Yes

Note

7 is a **compound proposition**

Note

We denote separate propositions as: p, q, r

1.1.2 Logical Operations (connectives)

Name	Notation	Phrasing	Truth Value
Disjunction (or)	$p \vee q$	"p or q"	True when p, q or both are TRUE
Conjunction (and)	$p \wedge q$	"p and q"	True when both are TRUE
Negation (not)	$\neg p$	"not p"	The opposite truth value of p
Exclusive Or (XOR)	$p \oplus q$	"p or q, but not both"	TRUE when only one is TRUE

1.3 Section 1.3 - Logical Equivalence

Recall 1 (Logical Equivalence)

Recall that two propositions are **logically equivalent** if they have the same truth table

1.3.1 Testing for Equivalence using Truth Tables

Example 2

Determine if the following are logically equivalent

1. $p \implies q$ and $\neg p \vee q$

Solution:

p	q	$p \implies q$	$\neg p$	$\neg p \vee q$
T	T	T	F	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

$$\therefore p \implies q \equiv \neg p \vee q$$

2. $\neg(p \wedge q)$ and $\neg p \vee \neg q$

Solution:

p	q	$p \wedge q$	$\neg(p \wedge q)$	$\neg p$	$\neg q$	$\neg p \vee \neg q$
T	T	T	F	F	F	F
T	F	F	T	F	T	T
F	T	F	T	T	F	T
F	F	F	T	T	T	T

$$\therefore \neg(p \wedge q) \equiv \neg p \vee \neg q$$

3. $(p \implies q) \vee (q \implies r)$ and $p \implies (q \vee r)$

Solution:

p	q	r	$p \implies r$	$q \implies r$	$(p \implies r) \vee (q \implies r)$	$q \vee r$	$p \implies (q \vee r)$
T	T	T	T	T	T	T	T
T	T	F	F	F	F	T	T
T	F	T	T	T	T	T	T
T	F	F	F	T	T	F	F
F	T	T	T	T	T	T	T
F	T	F	T	F	T	T	T
F	F	T	T	T	T	T	T
F	F	F	T	T	T	F	T

$$\therefore (p \implies q) \vee (q \implies r) \not\equiv p \implies (q \vee r)$$

So far we know the following:

- $p \implies q \equiv \neg p \vee q$
- DeMorgan's Law
 - $\neg(p \vee q) \equiv \neg p \wedge \neg q$
 - $\neg(p \wedge q) \equiv \neg p \vee \neg q$
- Commutative (swap places)
 - $p \vee q \equiv q \vee p$
 - $p \wedge q \equiv q \wedge p$
- Associative (swap grouping)
 - $(p \vee q) \vee r \equiv p \vee (q \vee r)$
 - $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$
- Distributive (distributing operations)
 - $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$
 - $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$
- Identity
 - $p \wedge T \equiv p$
 - $p \vee F \equiv p$
- Domination Laws
 - $p \vee T \equiv T$
 - $p \wedge F \equiv F$
- Impotent Laws
 - $p \vee p \equiv p$

$$- p \wedge p \equiv p$$

Definition 2 (Tautology and Contradiction)

A proposition that is always **TRUE** is a **tautology**

A proposition that is always **FALSE** is a **contradiction**

Note

When proving tautologies or contradictions, you could either:

1. Use Truth Tables
2. Use the Laws from earlier to reduce a compound proposition into **TRUE** or **FALSE**

But it's advised to use Truth Tables, since it's easy to get lost in your work when using the Laws

Also, ALWAYS start by **converting conditionals**

Example 3 (Examples of Tautologies and Contradictions)

Determine if the following are tautologies, contradictions, or neither

$$1. (p \wedge q) \implies q$$

Proving with Truth Tables

p	q	$p \wedge q$	q	$(p \wedge q) \implies q$
T	T	T	T	T
T	F	F	F	T
F	T	F	T	T
F	F	F	F	T

$$\therefore (p \wedge q) \implies q \equiv \text{tautology}$$

Proving with Logical Equations

$$\begin{aligned}
 (p \wedge q) \implies q &\equiv \neg(p \wedge q) \vee q && \text{(Convert Conditional)} \\
 &\equiv (\neg p \vee \neg q) \vee q && \text{(DeMorgan's Law)} \\
 &\equiv \neg p \vee (\neg q \vee q) && \text{(Associative)} \\
 &\equiv \neg p \vee T && \\
 &\equiv T && \text{(Domination)}
 \end{aligned}$$

$$\therefore (p \wedge q) \implies q \equiv \text{tautology}$$

2. There's another example but I'm too lazy

Note

When proving the **logical equivalence** of two propositions, start with the **MORE COMPLICATED** proposition

Example 4 (Proving Logical Equivalence of Two Propositions)

Determine if $(p \vee q) \implies r \equiv (p \implies r) \wedge (q \implies r)$

$(p \implies r) \wedge (q \implies r)$ seems more complicated, so we'll start with that

$$\begin{aligned}(p \implies r) \wedge (q \implies r) &\equiv (\neg p \vee r) \wedge (\neg q \vee r) && \text{(Convert Conditionals)} \\ &\equiv (\neg p \wedge \neg q) \vee r && \text{(Un-Distribute)} \\ &\equiv \neg(p \vee q) \vee r && \text{(DeMorgan's Law)} \\ &\equiv (p \vee q) \implies r && \text{(Convert back to Conditional)}\end{aligned}$$

$$\therefore (p \vee q) \implies r \equiv (p \implies r) \wedge (q \implies r)$$

1.4 Section 1.4 - Propositional Functions and Quantifiers

First, let's start with an example

Example 5

Is $x + 3 = 10$ a proposition?

Solution:

No, since x is not known

However, if x is known, then $x + 3 = 10$ is a proposition

The truth value depends on the value of x

Definition 3 (Propositional Function)

We call $x + 3 = 10$ a **propositional function** since the truth value of the proposition depends on the value of x

Example 6 (Examples of Propositional Functions)

For the given propositional functions, determine the truth value for the given value of variable(s)

1. $P(x) : x + 3 = 10$

(a) $P(6)$

Solution:

FALSE since $6 + 3 \neq 10$

(b) $P(7)$

Solution:

TRUE since $7 + 3 = 10$

2. $P(a, b, c) : a^2 + b^2 = c^2$

(a) $P(11, 60, 61)$

Solution:

TRUE

$$11^2 + 60^2 = 121 + 3600 = 3721$$

$$61^2 = (60 + 1)^2 = 3600 + 120 + 1 = 3721$$

(b) $P(147, 201, 513)$

Solution:

FALSE

$$147^2 + 201^2 = \text{odd} + \text{odd} = \text{even}$$

$$513^2 = \text{odd} \neq \text{even}$$

Example 7 (Examples of Notation)

1. Is "there exists a real number x such that $x + 3 = 10$?" a proposition?

Solution:

Yes, and this proposition is **TRUE** since $x = 7$ works

We can also rewrite this as:

$$\exists x \in \mathbb{R} \mid x + 3 = 10$$

2. Is "for all real numbers x , $x + 3 = 10$ " a proposition?

Solution:

Yes, but the proposition is **FALSE** since $x = 6$ doesn't work

We can also rewrite this as

$$\forall x \in \mathbb{R} \mid x + 3 = 10$$

Definition 4 (Quantifiers)

The **existential quantifier** $\exists$ means "there exists"

The **universal quantifier** $\forall$ means "for all" or "for every"

Example 8 (Examples of Quantifiers)

Write each proposition using quantifiers, and determine their truth values

1. There exists an integer x such that $x^2 + 2 > 3x$

Solution:

$$\exists x \in \mathbb{Z} \mid x^2 + 2 > 3x$$

TRUE since $x = 3$ gives $9 + 2 > 9$

2. For all real numbers x , $x^2 + 2 > 3x$

Solution:

$$\forall x \in \mathbb{R} \mid x^2 + 2 > 3x$$

FALSE since $x = 1$ gives $1 + 2 > 3$

Note

We only need one counter example to disprove a proposition that has $\forall$

3. There exists an integer x such that $x^3 = 31687$

Solution:

$$\exists x \in \mathbb{Z} \mid x^3 = 31687$$

FALSE since $\sqrt[3]{31687} \notin \mathbb{Z}$

Note

We needed to verify that every integer does not satisfy the proposition in order to prove it

The negations of each quantifier are as follows:

$$\neg \forall \equiv \exists$$

$$\neg \exists \equiv \forall$$

Example 9 (Negation of Quantifiers)

Give the negation of the following propositions:

1. $\forall x \in \mathbb{R} \mid x^2 + 1 > 0$ **TRUE**

Solution:

$$\exists x \in \mathbb{R} \mid x^2 + 1 \leq 0$$
 FALSE

2. $\exists a \in \mathbb{Z} \mid a^3 + 2 = 3a$ **TRUE** ($a = 1$)

Solution:

$$\forall a \in \mathbb{Z} \mid a^3 + 2 \neq 3a$$
 FALSE

3. For every real number x , $x^2 + 30 < 11x$ **FALSE** ($x = 1$)

Solution:

$$\exists x \in \mathbb{R} \mid x^2 + 30 \geq 11x$$
 TRUE

4. For every prime integer $p > 3$, $p^2 - 1$ is divisible by 24 **TRUE**

Solution:

There exists a prime integer $p > 3$, $p^2 - 1$ is not divisible by 24 **FALSE**

Work

$$p^2 - 1$$

We want $p^2 - 1$ divisible by both 8 and 3

$$(p^2 - 1) = (p - 1)(p + 1)$$

$$(p - 1)(p + 1) = (\text{even})(\text{even})$$

- p is prime
- $p > 3$
- p is odd

Example 10

Statement	When TRUE	When FALSE
$\forall x, P(x)$	$P(x)$ is TRUE for all x	There exists at least one x such that $P(x)$ is FALSE
$\exists x, P(x)$	There exists at least one x such that $P(x)$ is TRUE	For all x such that $P(x)$ is FALSE

Example 11 (Examples with Negation)

Statement	Equivalent	When TRUE	When FALSE
$\neg \forall x, P(x)$	$\exists x, \neg P(x)$	There is at least one x such that $P(x)$ is FALSE	For all x , $P(x)$ is TRUE
$\neg \exists x, P(x)$	$\forall x, \neg P(x)$	$P(x)$ is FALSE for all x	There exists at least one x such that $P(x)$ is TRUE

Example 12 (Examples of Rewriting with Quantifiers)

Rewrite the following with Quantifiers

1. For all positive integers, either $6n + 1$ or $6n - 1$ is prime

Solution:

$\forall n \in \mathbb{Z}^+, 6n - 1$ is prime $\vee 6n + 1$ is prime

FALSE since if $n = 20$ then $6n - 1 = 119$ and $6n + 1 = 121$ are **composite** (not prime)

2. There exists an integer n such that $n^2 + n + 41$ is composite

Solution:

$\exists n \in \mathbb{Z}, n^2 + n + 41$ is composite

TRUE since if $n = 41$ then

$$\begin{aligned} n^2 + n + 41 &= 41^2 + 41 + 41 \\ &= 41(41 + 1 + 1) \end{aligned}$$

which is composite

1.5 Section 1.5 - Nested Quantifiers

Let $\mathbb{P}$ be the set of prime integers

$$\mathbb{P} = \{2, 3, 5, 7, 11, \dots\}$$

With this in mind, we can write the following in two ways:

Consider the statement "for all positive integers n , $6n - 1$ or $6n + 1$ is prime"

1. $\forall n \in \mathbb{Z}^+, 6n - 1 \in \mathbb{P} \vee 6n + 1 \in \mathbb{P}$
2. $\forall n \in \mathbb{Z}^+ \exists p \in \mathbb{P}, 6n - 1 = p \vee 6n + 1 = p$

Example 13 (Example of Nested Quantifiers) 1. For all real x there exists a real number y such that $xy = 1$

$$\forall x \in \mathbb{R} \exists y \in \mathbb{R}, xy = 1$$

FALSE, since if $x = 0$, then $xy = 0 \neq 1$

2. For all positive real x , there exists a real number y such that $xy = 1$

$$\forall x \in \mathbb{R}^+ \exists y \in \mathbb{R}, xy = 1$$

TRUE since if $x \in \mathbb{R}^+$, then $x > 0$, and if $y = \frac{1}{x}$, then $xy = x \frac{1}{x} = 1$

Note

Given x , we can find y

y depends on x

But what happens if we switch $\forall$ and $\exists$?

3. $\exists y \in \mathbb{R} \forall x \in \mathbb{R}^+, xy = 1$

FALSE since if $x = 1$, then $y = 1$, but if $x = 2$, then $y = \frac{1}{2}$, and $1 \neq \frac{1}{2}$

Note

There is a single y that works **FOR ALL** x

y does not depend on x

4. For all positive integers a and b , we have $ab > 0$

$$\forall a \in \mathbb{Z}^+ \forall b \in \mathbb{Z}^+, ab > 0$$

Note

This is the same as $\forall a, b \in \mathbb{Z}^+$

TRUE since $a, b \in \mathbb{Z}^+$, then $a > 0$ and $b > 0$, hence $ab > 0$

5. There exists integers a , and b such that $a^2 + b^2 = 464$

$$\exists a \in \mathbb{Z} \exists b \in \mathbb{Z}, a^2 + b^2 = 464$$

$$\exists b \in \mathbb{Z} \exists a \in \mathbb{Z}, a^2 + b^2 = 464$$

TRUE $a = 8, b = 20,$

$$\begin{aligned} a^2 + b^2 &= 8^2 + 20^2 \\ &= 64 + 400 \\ &= 464 \end{aligned}$$

Note

The order of the **SAME** quantifier doesn't matter, only for **DIFFERENT** quantifiers does it matter

Example 14

Rewrite the following using Quantifiers

1. "There exists positive integers a and b such that $a^2 + b^2 = 15625$ "

(a) Write as Quantifier

$$\exists a \in \mathbb{Z}^+ \exists b \in \mathbb{Z}^+, a^2 + b^2 = 15625$$

(b) Write the negation

$$\forall a \in \mathbb{Z}^+ \forall b \in \mathbb{Z}^+, a^2 + b^2 \neq 15625$$

Work

$$a^2 + b^2 = 5^6$$

$$a^2 + b^2 = 5^4 \cdot 5^2$$

$$3^2 + 4^2 = 5^2$$

$$5^4(3^2 + 4^2) = 5^4 \cdot 5^2$$

$$5^4 \cdot 3^2 + 5^4 \cdot 4^2 = 5^6$$

$$(5^2 \cdot 3)^2 + (5^2 \cdot 4)^2 = 5^6$$

$$(75)^2 + (100)^2 = 5^6$$

$$\therefore a = 75, b = 100$$

TRUE

There's more examples but I'm just too lazy to write them if you get the chance, they're on page 3 of your notes

1.7 Section 1.7 Intro to Proofs

What do we prove? Usually that some pattern or formula is true.

Example 15

$$\begin{aligned}
6 + 8 &= 14 \\
12 + 16 &= 28 \\
2 + 2 &= 4
\end{aligned}$$

We believe that the sum of 2 even integers is even.

Try to prove that: even + even = even

Work An even number can be defined as $2n$, where n is an integer

$$2n + 2m = 2(n + m)$$

Proof. let a, b be even integers.

Then there exists integers n, m such that $a = 2n, b = 2m$

$$a + b = 2n + 2m = 2(m + n)$$

is even, as desired

□

But writing it this way is too long. Let's define some notation:

1.7.1 Set Notation

Name	Symbol	Set Notation
Set of all natural numbers	$\mathbb{N}$	$\{0, 1, 2, 3, \dots\}$
Set of all integers	$\mathbb{Z}$	$\{0, -1, 1, -2, 2, \dots\}$
Set of positive integers	$\mathbb{Z}^+$	$\{1, 2, 3, \dots\}$
Set of negative integers	$\mathbb{Z}^-$	$\{-1, -2, -3, \dots\}$
Set of rational numbers	$\mathbb{Q}$	$\{\frac{a}{b} \mid a, b \in \mathbb{Z}, b \neq 0\}$
Set of real numbers	$\mathbb{R}$	Limit of all convergent sequences of real numbers
Set of all complex numbers	$\mathbb{C}$	$\{a + bi \mid a, b \in \mathbb{R}, i^2 = -1\}$

$\exists$ means there exists

Let's rewrite the previous example using this new notation

$a, b \in \mathbb{Z}$, both even

Then, $\exists m, n \in \mathbb{Z}$ such that

$$a = 2m, b = 2n$$

Their sum $a + b = 2m + 2n = 2(m + n)$

is even, as desired

1.7.2 Methods of Proving $p \implies q$

1. **Direct Proof:**

Start with p and *math* then conclude q

2. **Indirect Proof:**

Prove the contrapositive proof $\neg q \implies \neg p$

3. **Finite Cases:**

Use the fact that $p \equiv p_1 \vee p_2 \vee \dots \vee p_n$

Then prove $(p_1 \implies q) \wedge (p_2 \implies q) \wedge \dots \wedge (p_n \implies q)$

Chapter 2

Section 2 - Structures: Sets, Functions, Sequences, Sums, and Matrices

2.1 Section 2.1 - The Joy of Sets

Definition 5 (Sets and Elements)

A **set** is an unordered collection of objects called **elements**

$$\mathbb{N} = \{1, 2, 3, 4, \dots\} \quad (\text{Natural Numbers})$$

$$\text{Whole Numbers} = \{0, 1, 2, 3, \dots\}$$

$$\mathbb{Z} = \{0, -1, 1, -2, 2, -3, 3, \dots\}$$

$$\mathbb{Q} = \left\{ \frac{a}{b} \mid a, b \in \mathbb{Z}, b \neq 0 \right\}$$

$$\mathbb{R} = \text{limit of all convergent sequences of rational numbers}$$

$$\mathbb{C} = \{a + bi \mid a, b \in \mathbb{R}, i^2 = -1\}$$

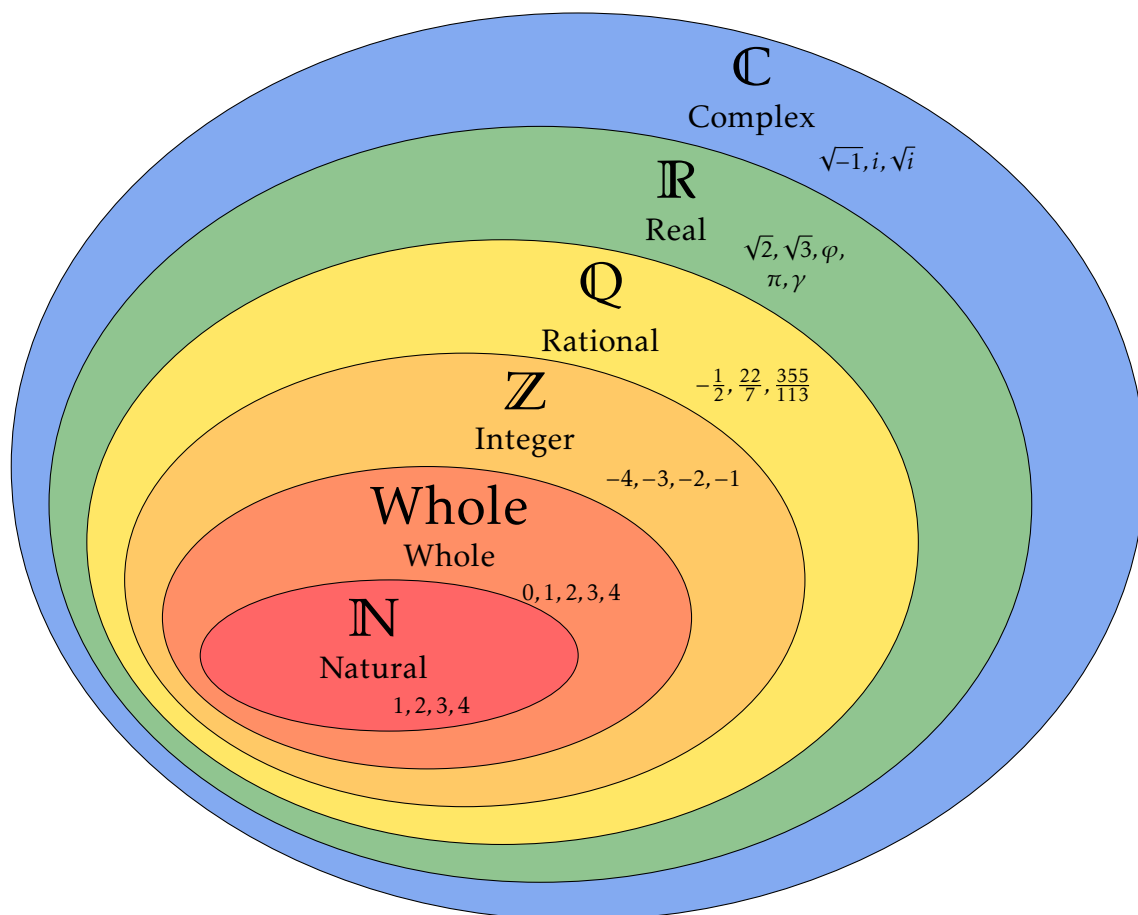
Note

This notation for $\mathbb{Q}$ is called **Set Builder Notation**

It's read as "Q is the set $\frac{a}{b}$ such that $a, b \in \mathbb{Z}, b \neq 0$ "

Note

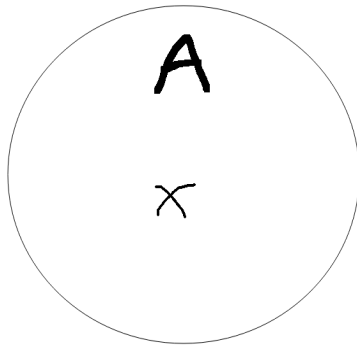
Notice that every natural number is a whole number. So we say "the set of natural numbers is a **subset** of the set of whole numbers"



Definition 6 (Elements in Sets)

If x is an element of the set A , then we write

$$x \in A$$



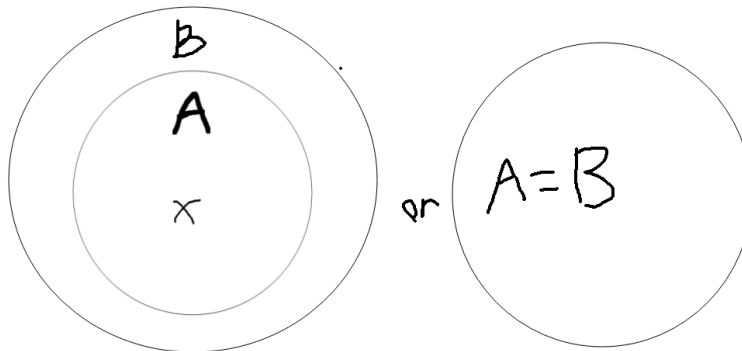
Definition 7 (Empty Set)

The set containing no elements is called the **empty set**, and is denoted by $\{\}$ or $\emptyset$

Given sets A and B such that for all $x \in A$ we have $x \in B$, then we say **A is a subset of B** and we write

$$A \subseteq B$$

Which can mean two things:



Definition 8 (Proper Sets)

If there exists $x \in B$ with $x \notin A$, then we say that A is a **proper subset** of B and we write

$$A \subsetneq B$$

Definition 9 (Cardinality)

Given a set A , the **cardinality** of a set is the number of elements in the set A , denoted by

$$|A|$$

Definition 10 (Power Set)

The **power set** of a given set S is the set of all subsets of S , denoted by

$$\mathcal{P}(S)$$

$$|\mathcal{P}(S)| = 2^{|S|}$$

Note

This includes the empty set, $\{\}$

Definition 11 (Cartesian Product)

Given sets A, B , the **cartesian product** is the set

$$A \times B = \{(a, b) \mid a \in A, b \in B\}$$

$$|A \times B| = |A| \cdot |B|$$

$$A^2 = A \times A$$

Example 16

Let $A = \{1, 2, 3\}, B = \{1, 2\}$

(a) Is $A \subseteq B$?

Solution:

No since $3 \in A, 3 \notin B$

(b) Is $B \in A$?

Solution:

Yes since for all $x \in B, x \in A$

(c) What is the power set of B ?

Solution:

$$\mathcal{P} = \{\{\}, \{1\}, \{2\}, \{1, 2\}\}$$

(d) What is the power set of A ?

Solution:

$$\mathcal{P}(A) = \{\{\}, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$$

Note

If A is a finite set (ie $|A|$ is finite), then $|\mathcal{P}(A)| = 2^{|A|}$

(e) Determine $A \times B$

Solution:

$$A \times B = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 1), (3, 2)\}$$

$$B \times A = \{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3)\}$$

$$A \times B \neq B \times A$$

$$|A \times B| = |B \times A| = |A| \cdot |B|$$

2.2 Section 2.2 - Set Operations

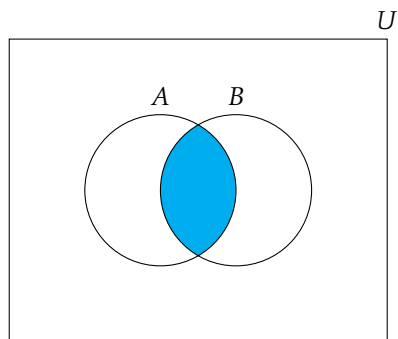
Let's consider a venn diagram which is a way of visualizing sets

Let U represent the universal set

Definition 12 (Intersection)

Elements that are in **both** A and B is called the intersection of A and B

$$A \cap B$$



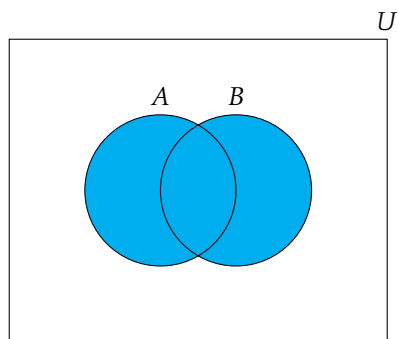
Definition 13 (Union)

Elements that are in **either** A or B is called the **union** of A and B

$$A \cup B$$

Also

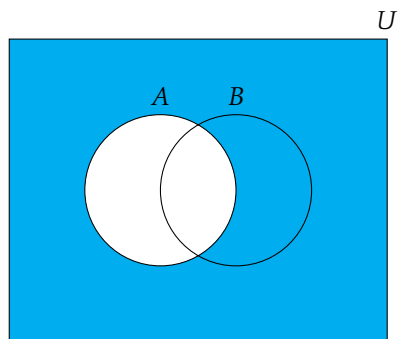
$$|A \cup B| = |A| + |B| - |A \cap B|$$



Definition 14 (Compliment)

Elements that just aren't in A is called the **compliment** of A

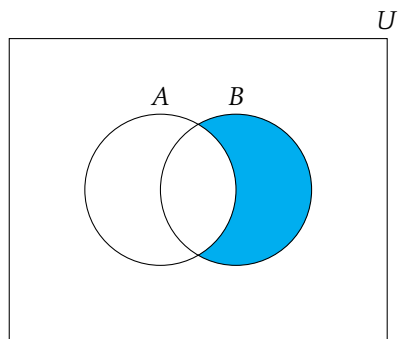
$$\bar{A}$$



Definition 15 (Difference)

Elements in B but not in A is called the **difference**

$$B - A$$



With sets, we have 3 operations:

1. Compliment $\bar{A}$ (analogous to $\neg$)
2. Union $A \cup B$ (analogous to $\vee$)
3. Intersection $A \cap B$ (analogous to $\wedge$)

Example 17

Let

$$U = \{x \in \mathbb{Z} \mid 0 \leq x \leq 20\}$$

$$A = \{x \in U \mid x \text{ is even}\}$$

$$B = \{x \in U \mid x \text{ is divisible by 3}\}$$

Determine the following

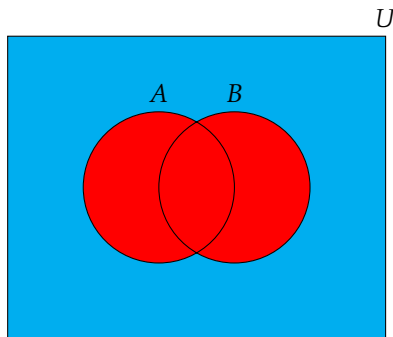
1. $A = \{0, 2, 4, 6, 8, 10, 12, 14, 16, 18, 20\}$
2. $B = \{0, 3, 6, 9, 12, 15, 18\}$
3. $A \cap B = \{0, 6, 12, 18\}$ (what they have in common)
4. $A \cup B = \{0, 2, 3, 4, 6, 8, 9, 10, 12, 14, 15, 16, 18, 20\}$ (whats in either one)
5. $\bar{A} = \{1, 3, 5, 7, 9, 11, 13, 15, 17, 19\}$ (whats not in A)
6. $B - A = \{3, 9, 15\}$ (what's in B thats not in A)
7. $|A| = 11$ (number of elements in A)
8. $|B| = 7$ (number of elements in B)
9. $|A \cup B| = 14$
10. $|A \cap B| = 4$ (number of elements that A and B have in common)
11. $|A| + |B| - |A \cap B| = 11 + 7 - 4 = 14 = |A \cup B|$

But how do these operations interact together?

Example 18

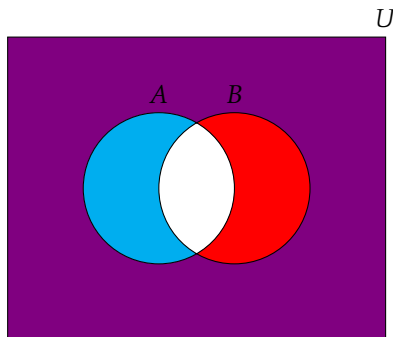
- $\overline{A \cup B}$

$$\begin{aligned} \text{Red} &= A \cup B \\ \text{Blue} &= \overline{A \cup B} \end{aligned}$$



- $\bar{A} \cap \bar{B}$

$$\begin{aligned} \text{Red} &= \bar{A} \\ \text{Blue} &= \bar{B} \\ \text{Purple} &= \bar{A} \cap \bar{B} \end{aligned}$$



$$\therefore \overline{A \cup B} = \overline{A} \cap \overline{B}$$

1. **DeMorgan's Laws**

- $\overline{A \cup B} = \overline{A} \cap \overline{B}$
- $\overline{A \cap B} = \overline{A} \cup \overline{B}$

2. **Distributive Laws**

- $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
- $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

3. **Complementation Law**

- $\overline{\overline{A}} = A$

4. **Commutative Law**

- $A \cup B = B \cup A$
- $A \cap B = B \cap A$

2.3 Section 2.3 - Functions

Definition 16

Let A and B be sets

A **relation** from A to B is a subset of $A \times B$

Example 19

Let $A = \{1, 4, 9\}$ and $B = \{\pm 1, \pm 2, \pm 3\}$

1. $|A \times B| = 3 \cdot 6 = 18$
2. Let R be the relation from A to B defined by

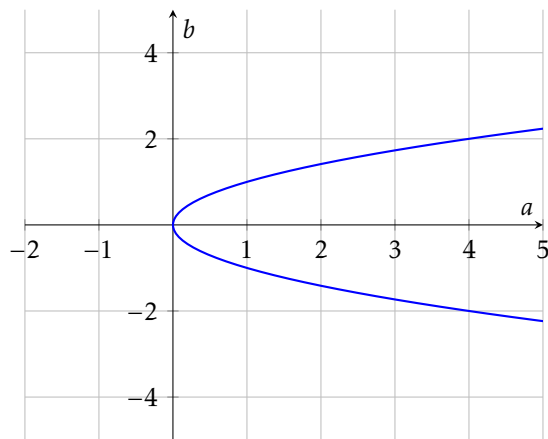
$$R = \{(a, b) \mid a \in A, b \in B, a = b^2\}$$

$$R = \{(1, 1), (1, -1), (4, 2), (4, -2), (9, 3), (9, -3)\}$$

Note

R is a relation but not a function

Why?



This graph of the relation fails the vertical line test, hence it is not a function

There exists an element of A that corresponds to more than one element of B

a	b
1	-1
1	1
4	-2
4	2
9	-3
9	3

Every function is a relation but not all relations are functions

Definition 17 (Function)

Let A and B be sets

A **function** f from A to B , denoted as

$$f : A \rightarrow B$$

is a relation from A to B such that

$$\forall a \in A \exists! b \in B, f(a) = b$$

$\exists!$ means "there exists a **unique**..."

Definition 18 (Domain and Codomain)

We call the set A the **domain** of f and the set B the **codomain**.

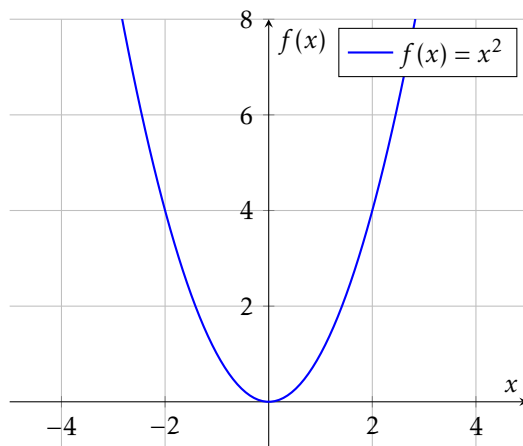
The **range** or **image** of f is the set

$$\text{Im}(f) = \{b \in B \mid \exists a \in A, f(a) = b\}$$

Example 20

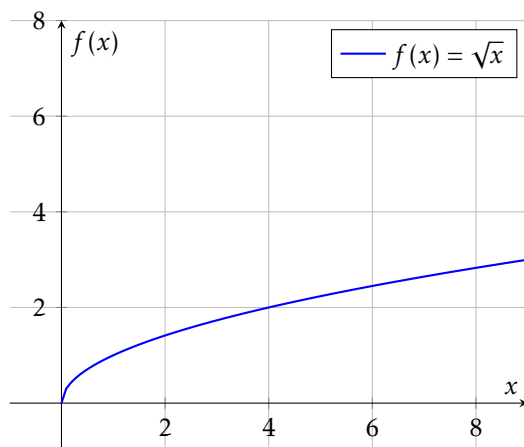
Determine if the following are functions

1. $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^2$



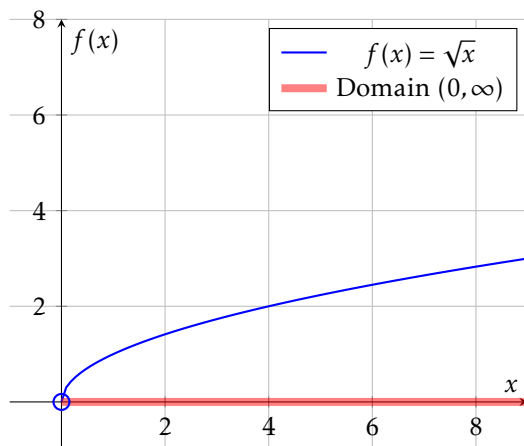
Domain: x axis
 Codomain: y axis
 $\text{Im}(f) = [0, \infty]$
 $\therefore$ Yes

2. $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = \sqrt{x}$



No since not all inputs are valid

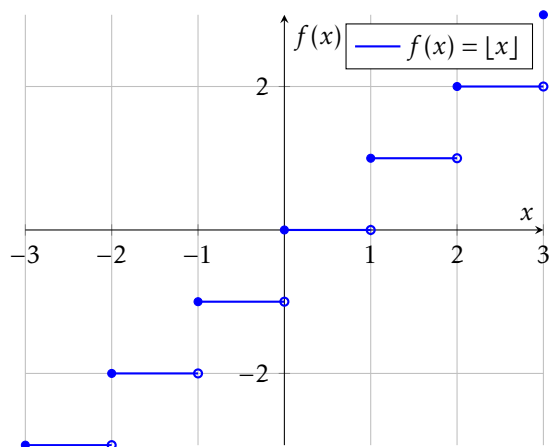
3. $f : \mathbb{R}^+ \rightarrow \mathbb{R}, f(x) = \sqrt{x}$



$$\text{Im}(f) = (0, \infty)$$

Yes since all the inputs have a unique output (the origin is excluded since the domain is $\mathbb{R}^+$)

4. $f : \mathbb{R} \rightarrow \mathbb{Z}, f(x) = \lfloor x \rfloor$ (floor function, aka greatest integer function)

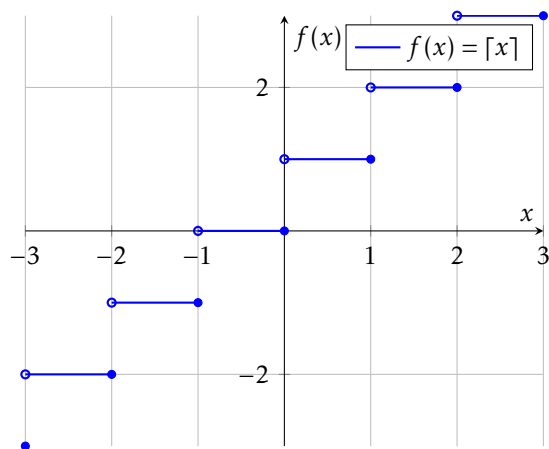


x	$\lfloor x \rfloor$
-1	-1
0	0
0.5	0
0.99	0
1.01	1

$$\text{Im}(f) = \mathbb{Z}$$

All of x axis is domain, but only **integer** parts of y axis is codomain

5. $f : \mathbb{R} \rightarrow \mathbb{Z}$, $f(x) = \lceil x \rceil$ (ceiling function, aka least integer function)



x	$\lfloor x \rfloor$
-1	-1
0	0
0.5	1
0.99	1
1.01	2

$$\text{Im}(f) = \mathbb{Z}$$

All of x axis is domain, but only **integer** parts of y axis is codomain

Definition 19 (Surjective)

Let $f : A \rightarrow B$ be a function in the set A and B

We say f is **onto** or **surjective** if

$$\text{Im}(f) = B$$

or

$$\forall b \in B \exists a \in A, f(a) = b$$

or

All of codomain is covered by the function

Definition 20 (Injective)

We say f is **one to one** or **injective** if

$$\forall b \in B, \text{ the equation } f(a) = b \text{ has at most one solution}$$

Another way of thinking of this is:

Injective if it passes the horizontal line test

or

there is a unique b for all a

or

no two inputs have the same output

or

$$f(a) = f(b) \implies a = b$$

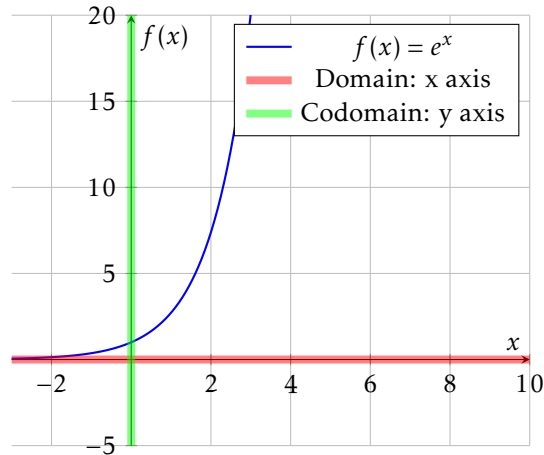
Definition 21 (Bijective)

If f is both **surjective** and **injective**, then we say f is **bijective**

Example 21

Determine if the following functions are injective, surjective, and / or bijective

1. $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = e^x$

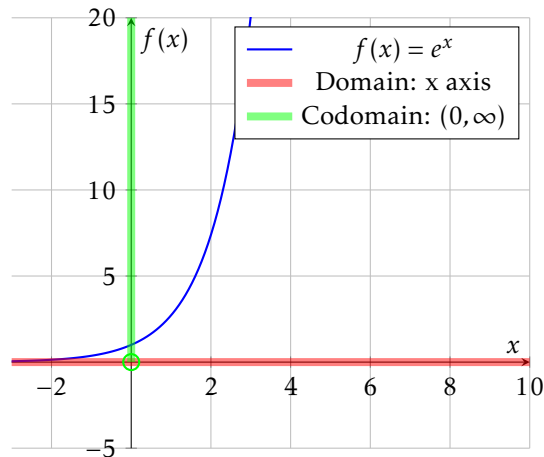


Injective since all x values have a valid output

Not surjective since not all values on the y axis have a corresponding input value

Not bijective since it's not surjective

2. $f : \mathbb{R} \rightarrow \mathbb{R}^+, f(x) = e^x$



Injective since all x values have a valid output

Surjective since all values on the y axis have a corresponding input value

Bijective since it's both surjective and injective

There's more examples, write them down when you get the chance

Definition 22 (Composition)

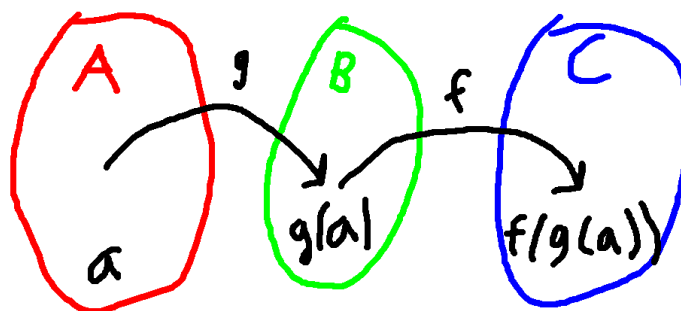
Given the functions $f : B \rightarrow C$ and $g : A \rightarrow B$

Where A, B, C are sets, then we define

$$f \circ g : A \rightarrow C \text{ by}$$

$$(f \circ g)(a) = f(g(a))$$

called the **composition** of f and g



Example 22 (Examples of Compositions)

Let $f(x) = x - 3$, $g(x) = x^3 + 9x^2 + 27x + 27 = (x + 3)^3$

Compute the following

1. $(f \circ g)(x)$

$$\begin{aligned} (f \circ g)(x) &= f(g(x)) \\ &= (x^3 + 9x^2 + 27x + 27) - 3 \\ &= x^3 + 9x^2 + 27x + 24 \end{aligned}$$

$$2. (g \circ f)(x)$$

$$\begin{aligned}(g \circ f)(x) &= g(f(x)) \\ &= (x-3)^3 + 9(x-3)^2 + 27(x-3) + 27 \\ &= [(x-3) + 3]^3 \\ &= x^3\end{aligned}$$

2.4 Section 2.4 - Sequences and Series

Definition 23 (Sequence, Explicit Formula / nth term)

A **sequence** is an ordered list of numbers

$$a_1, a_2, a_3, \dots$$

Equivalently, a sequence is a function $f: \mathbb{Z}^+ \rightarrow \mathbb{R}$

We call f the **explicit formula** or **nth term** of a sequence

Example 23

Determine the next 3 terms of the following sequences and the explicit formula

$$1. \quad 5 \xrightarrow{+7}, 12 \xrightarrow{+7}, 19 \xrightarrow{+7}, 26 \xrightarrow{+7}, \dots, 33 \xrightarrow{+7}, 44 \xrightarrow{+7}$$

Solution:

$$\begin{aligned}a_n &= 7n + b \\ 5 &= a_1 = 7(1) + b \\ b &= 7 \\ \boxed{a_n &= 7n - 2}\end{aligned}$$

$$2. \quad 7 \xrightarrow{\cdot 3}, 21 \xrightarrow{\cdot 3}, 63 \xrightarrow{\cdot 3}, 189 \xrightarrow{\cdot 3}, \dots, 567 \xrightarrow{\cdot 3}, 1701 \xrightarrow{\cdot 3}, 5103 \xrightarrow{\cdot 3}$$

Solution:

$$a_n = c3^n$$

$$a_1 = 7 = c3^1$$

$$c = \frac{7}{3}$$

$$a_n = \frac{7}{3}3^n$$

$$\boxed{a_n = 7 \cdot 3^{n-1}}$$

3. 1, 2, 6, 24, 120, 720, 5040

$\boxed{a_n = n!}$, read as "n factorial"

$$0! = 1$$

4. 1, 1, $\underbrace{2}_{1+1}$, $\underbrace{3}_{2+1}$, $\underbrace{5}_{3+2}$, $\underbrace{8}_{5+3}$, $\underbrace{13}_{8+5}$, $\underbrace{21}_{13+8}$,

Solution:

$$f_n = \frac{1}{\sqrt{5}} \left[\left(\frac{1+\sqrt{5}}{2} \right)^n - \left(\frac{1-\sqrt{5}}{2} \right)^n \right] \quad (\text{We'll derive this later})$$

Another solution is

$$a_1 = 1$$

$$a_2 = 2$$

$$a_n = a_{n-1} + a_{n-2}, n \geq 3$$

Definition 24

A **recursive** sequence is defined in terms of the previous terms of the sequence

Example 24

Here are some examples of recursive sequences

$$1. \ a_1 = 2, a_2 = 3, a_n + 1 = 5a_n - 6a_{n-1}, n \geq 2$$

$$\begin{aligned}
 n = 2, a_3 &= 5a_2 - 6a_1 = 5 \cdot 3 - 6 \cdot 2 = 15 - 12 = 3 \\
 n = 3, a_4 &= 5a_3 - 6a_2 = 5 \cdot 3 - 6 \cdot 3 = 15 - 18 = -3 \\
 n = 4, a_5 &= 5a_4 - 6a_3 = 5 \cdot (-3) - 6 \cdot 3 = -15 - 18 = -33
 \end{aligned}$$

$$2. \ a_1 = 1, a_2 = 2, a_3 = 4, a_n = a_{n-1} + a_{n-2} - n - 3, n \geq 4$$

$$a_4 = 4 + 2 - 1 = 5$$

$$a_5 = 5 + 4 - 2 = 7$$

$$a_6 = 7 + 5 - 4 = 8$$

Definition 25 (Series)

A **series** is the sum of an ordered list of numbers

$$a_1 + a_2 + a_3 + \dots$$

$$\sum_{k=1}^{\infty} a_k$$

Where:

k = Index

1 = Starting value

∞ = Ending value

$a_k = k^{\text{th}}$ term

Example 25

Find the sum of the series

$$1 + 2 + 3 + \dots + 100$$

Solution:

$$\begin{aligned} &1 + 2 + 3 + \dots + 98 + 99 + 100 \\ &(1 + 100) + (2 + 99) + (3 + 98) + \dots \\ &(101) + (101) + (101) + \dots \end{aligned}$$

Since there's 100 things we need to add, there's 50 pairs

So

$$\begin{aligned} \text{sum} &= 101(50) \\ &= 5050 \end{aligned}$$

Example 26

But what about in general?

$$1 + 2 + 3 + \dots + n = ?$$

We have two cases:

1. When n is even
2. When n is odd

If we assume n is even, then

$$\boxed{1 + 2 + 3 + \dots + n = (n + 1) \cdot \frac{n}{2}} \quad (2.1)$$

Since $(n + 1)$ gives the value of each pair and $\frac{n}{2}$ gives the number of pairs

If we assume n is odd, then

$$\begin{aligned} [1 + 2 + 3 + \dots + (n - 1)] + n &= n \cdot \frac{n - 1}{2} \\ &= \frac{n^2 + n}{2} \\ &= (n + 1) \cdot \frac{n}{2} \end{aligned}$$

since n is the value of each pair $(n - 1 + 1)$ and $\frac{n-1}{2}$ is the number of pairs

Another way to think about it is to let S represent the sum

$$\begin{array}{r}
 S = 1 + 2 + 3 + \dots + (n-2) + (n-1) + n \\
 S = n + (n-1) + (n-2) + \dots + 3 + 2 + 1 \quad \text{(Switch Order)} \\
 \hline
 2S = \underbrace{(n+1) + (n+1) + (n+1) + \dots + (n+1) + (n+1) + (n+1)}_n \quad \text{(Add)} \\
 \boxed{S = (n+1) \cdot \frac{n}{2}}
 \end{array}$$

Writing this is summation notation

$$\sum_{k=1}^n \frac{n(n+1)}{2}$$

To Review...

1.

$$\sum_{k=1}^n c = \underbrace{c + c + \dots + c}_n = \boxed{nc} \quad \text{(North Carolina Rule)}$$

2.

$$\sum_{k=1}^n k = 1 + 2 + 3 + \dots + n = \boxed{(n+1)\frac{n}{2}} \quad \text{(Pairing Rule)}$$

3.

$$\sum_{k=1}^n k^2 = 1^2 + 2^2 + 3^2 + \dots + n^2 = \boxed{\frac{n(n+1)(2n+1)}{6}} \quad \text{(Prove Later)}$$

4.

$$\sum_{k=1}^n k^3 = 1^3 + 2^3 + 3^3 + \dots + n^3 = \boxed{\left[\frac{n(n+1)}{2}\right]^2} \quad \text{(Prove Later)}$$

You **MUST** know these 4! They will be on exams!

Example 27

Compute the Following

1.

$$\sum_{k=1}^6 k = \frac{6}{2}(6+1) = \boxed{21} \quad (\text{Use formula})$$

2.

$$\begin{aligned} \sum_{k=1}^6 2k &= (2 \cdot 1) + (2 \cdot 2) + (2 \cdot 3) + (2 \cdot 4) + (2 \cdot 5) + (2 \cdot 6) \\ &= 2(1 + 2 + 3 + 4 + 5 + 6) \\ &= 2 \sum_{k=1}^6 k \\ &= 2 \cdot 21 \\ &= \boxed{42} \end{aligned}$$

NoteThis means that we can **factor** in sums!

3.

$$\sum_{k=1}^6 2 = 6 \cdot 2 = \boxed{12} \quad (\text{Use formula})$$

4.

$$\begin{aligned} \sum_{k=1}^6 (k+2) &= (1+2) + (2+2) + (3+2) + (4+2) + (5+2) + (6+2) \\ &= (1+2+3+4+5+6) + (2+2+2+2+2+2) \\ &= \sum_{k=1}^6 k + \sum_{k=1}^6 2 \\ &= 21 + 12 \\ &= \boxed{33} \end{aligned}$$

Note

This means that in general,

$$\sum_{k=1}^n c a_k = c \sum_{k=1}^n a_k \quad (\text{Factor out constants})$$

$$\sum_{k=1}^n (a_k + b_k) = \sum_{k=1}^n a_k + \sum_{k=1}^n b_k \quad (\text{Split})$$

5.

$$\sum_{k=1}^{200} (k+1)^2 = \sum_{k=1}^{200} (k^2 + 2k + 1) \quad (\text{Expand})$$

$$= \sum_{k=1}^{200} k^2 + 2 \sum_{k=1}^{200} k + \sum_{k=1}^{200} 1 \quad (\text{Split up})$$

$$= \frac{200(201)(401)}{6} + 2 \cdot \frac{200(201)}{2} + 200 \cdot 1 \quad (\text{Use formulas})$$

$$\boxed{= 2727100} \quad (\text{Use Calculator})$$

We could also do

$$\sum_{k=1}^{200} (k+1)^2 = 2^2 + 3^2 + 4^2 + \dots + 201^2$$

$$= (1^2 + 2^2 + 3^2 + 4^2 + \dots + 201^2) - 1^2$$

$$= \frac{201(202)(403)}{2} - 1$$

$$\boxed{= 2727100} \quad (\text{Use Calculator})$$

Theorem 1 (Constant Shift)

This means that when you have a **constant shift** like $(k+1)$ is a shift from k , then

$$\sum_{k=1}^n f(k+a) = \sum_{k=1}^{n+a} f(k) - \sum_{k=1}^a f(k)$$

or

$$\sum_{k=m}^n a_k = \sum_{k=1}^n a_k - \sum_{k=1}^{m-1} a_k$$

Where $f(k)$ is a function like k^2 that has it's own formula for the sum like

$$\sum_{k=1}^n f(k) = \sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$$

Using this on the other example,

$$\begin{aligned} \sum_{k=1}^{200} (k+1)^2 &= \sum_{k=1}^{201} k^2 - \sum_{k=1}^1 k^2 \\ &= \frac{201(202)(403)}{2} - \frac{1(2)(1)}{2} \\ &= 2727101 - 1 \\ &= 2727100 \end{aligned} \quad \text{(Use Calculator)}$$

6.

$$\sum_{k=1}^{10} 2^k = S$$

$$S = 2^1 + 2^2 + 2^3 + \dots + 2^9 + 2^{10}$$

$$2S = 2^2 + 2^3 + 2^4 + \dots + 2^{10} + 2^{11}$$

(Subtract)

$$S - 2S = 2^1 - 2^{11}$$

$$S(1 - 2) = 2^1 - 2^{11}$$

$$S = \frac{2 - 2^{11}}{1 - 2} = \frac{2 - 2048}{-1}$$

$$\boxed{= 2046}$$

Note

This means that

$$S = \sum_{k=0}^n r^k$$

(When k starts at 0)

$$S = 1 + r^1 + r^2 + \dots + r^{n-1} + r^n$$

$$rS = 0 + r^1 + r^2 + r^3 + \dots + r^{n-1} + r^n + r^{n+1}$$

(Subtract)

$$S - rS = 1 - r^{n+1}$$

$$S(1 - r) = 1 - r^{n+1}$$

$$\boxed{S = \frac{1 - r^{n+1}}{1 - r}}$$

$$S = \sum_{k=1}^n r^k \quad (\text{When } k \text{ starts at } 1)$$

$$\begin{aligned} S &= r^1 + r^2 + \dots + r^{n-1} + r^n \\ &= r(1 + r^1 + r^2 + \dots + r^{n-2} + r^{n-1}) \\ &= r \sum_{k=1}^{n-1} r^k \\ &= r \cdot \frac{1 - r^{n-1+1}}{1 - r} \\ &= r \cdot \frac{1 - r^n}{1 - r} \end{aligned}$$

In general,

$$\sum_{k=\text{start}}^n r^k = \frac{r^{\text{start}} - r^{n+1}}{1 - r}$$

Note

This works for $r \in \mathbb{R}, (-\infty, \infty)$, including **negative** and **fractions**!

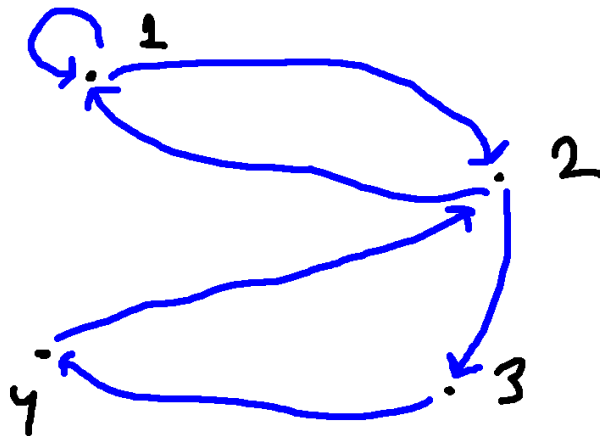
This can also work for sums going to ∞ , remember that a fraction to ∞ is just 0

There's more examples, but I'm too lazy to write them, so I'll just write the equations

Note

$$\sum_{k=0}^{\infty} k r^k = \sum_{k=1}^{\infty} k r^k = \boxed{\frac{r}{(1-r)^2}}$$

2.6 Section 2.6 - Matrices



Definition 26 (Directed Graph)

This is a **directed graph** and could be used to model:

- How webpages are linked
- Flights between cities
- Who knows who

How can a computer store this directed graph?

Definition 27 (Matrix)

We will use a **matrix**, which is a rectangular array of numbers

For this network, we define a matrix

$$A = [a_{ij}]$$

Where

$$a_{ij} = \begin{cases} 0 & \text{if web page } i \text{ doesn't link to } j \\ 1 & \text{if web page } i \text{ does link to } j \end{cases}$$

i = row (starts at 1)
 j = column (starts at 1)

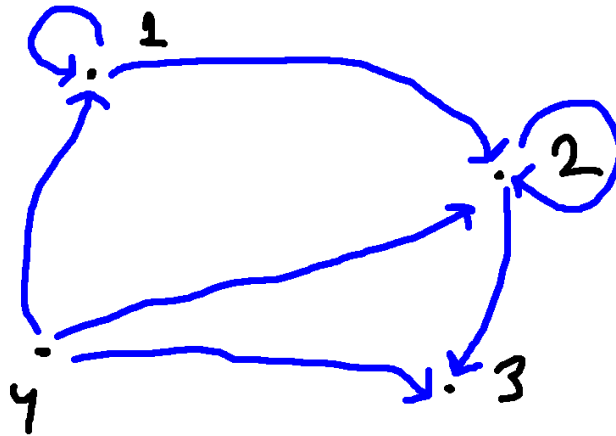
The resulting matrix would look like

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix}$$

For this network, the matrix is

$$A = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

Let's modify the graph now



This matrix is

$$B = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$

Matrix Addition and Subtraction

We define the addition of matrices by adding corresponding components

$$A + B = \begin{bmatrix} 2 & 2 & 0 & 0 \\ 1 & 1 & 2 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

The same can be said for Subtraction

$$A - B = \begin{bmatrix} 0 & 0 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 0 & 0 & 0 & 1 \\ -1 & 1 & -1 & 1 \end{bmatrix}$$

Note

Two matrices can only be added/subtracted if they're the same size

Question

Going back to the first network, how many ways can we start at 1 and return to 1 after passing through exactly 1 webpage?

$$1 \rightarrow 1 \rightarrow 1$$

$$1 \rightarrow 2 \rightarrow 1$$

$$a_{12} \rightarrow a_{21}$$

$$\begin{bmatrix} a_{11} & a_{12} & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} a_{11} & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ a_{21} & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

Example 28

Let

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}, B = \begin{bmatrix} 1 & 0 \\ -1 & 1 \\ 0 & 1 \end{bmatrix}, C = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

1. $A + B$

Solution:

Undefined, due to dimensions

2. $6A$

Solution:

$$6A = \begin{bmatrix} 6 & 12 & 18 \\ 24 & 30 & 36 \end{bmatrix}$$

3. AB

Solution:

$$AB = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 5 \\ -1 & 11 \end{bmatrix}$$

4. BA

Solution:

$$BA = \begin{bmatrix} 1 & 0 \\ -1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 3 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

5. BC

Solution:

$$BC = \begin{bmatrix} 1 & 0 \\ -1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 1 & 2 \\ 2 & 4 \end{bmatrix}$$

6. CB

Solution:

Undefined, due to dimensions

7. $C^2 = CC$

$$C^2 = CC = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} 5 & 10 \\ 10 & 20 \end{bmatrix} \\ = 5C$$

8. C^{11}

$$C^2 = 5C$$

$$C^3 = C^2 \cdot C$$

$$= 5C \cdot C$$

$$= 5C^2$$

$$= 5^2 C$$

$$C^{11} = 5^{10} C = 5^{10} \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

Okay let's now define some properties

2.6.1 Properties of Matrices

Definition 28 (Additive Identity)

We define the **Additive Identity** for matrices as

$$A_{n \times m} + \vec{0} = A_{n \times m}, \vec{0} = \begin{bmatrix} 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{bmatrix}$$

Definition 29 (Multiplicative Identity)

We define the **Multiplicative Identity** for matrices as

$${}_{n \times m} A \cdot I_n = A, I_n = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}$$

But A must be **square**!

Some matrices have a multiplicative inverse such that

Definition 30 (Multiplicative Inverse)

We define the **Multiplicative Inverse** of matrices as

$$A \cdot B = I$$

But only **some** matrices have them, not all do!

Also, A must be square to even qualify for possibly having one

For this class, we define

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Example 29

We have

$$\begin{bmatrix} 4 & 3 \\ 7 & 5 \end{bmatrix}^{-1} = \frac{1}{20 - 21} \begin{bmatrix} 5 & -3 \\ -7 & 4 \end{bmatrix}$$